

Lesson Overview: Data Lineage

In today's data-centric environment, the ability to **manage, govern, and ensure the quality of data is critical**. Effective data management practices are fundamental to maintaining **data integrity**, supporting **accurate analytics**, and enabling efficient **machine learning workflows**. As organisations increasingly rely on data to drive decisions, the skills to implement robust **data governance frameworks and tools** are essential for safeguarding data accessibility and trustworthiness.

In this lesson, we'll gain a comprehensive understanding of **data management and governance practices crucial for ensuring data quality and accessibility**. We'll learn about the roles of data catalogues, dictionaries, and lineage in safeguarding data integrity, and discover how feature stores contribute to efficient machine-learning workflows.



What You'll Learn:



Principles of data management and governance

Ensure high data quality by applying key data management and governance principles that support consistency and integrity.



Role of data catalogues

Understand how data catalogues make data assets easier to find and manage as centralised metadata repositories.



Data dictionaries and lineage

Explore data dictionaries and lineage for setting data standards and ensuring data traceability within governance.



Importance of feature stores

Discover how feature stores support machine learning projects by providing a reusable, scalable data repository.



Data lineage best practices and tools

Identify best practices for implementing data lineage and the tools that enhance data tracking and transparency.

Ready to dive in?

Reflect:

Think about a recent time when you relied on information to make an important decision — whether it was comparing reviews before buying a product, looking up health-related advice, or deciding where to travel. How did you know you could trust the information?

Consider how you might verify the quality, accuracy, or completeness of data in everyday scenarios. Now, think about why businesses need to be even more careful when making decisions based on their data.